

STANA DE VALE, Romania

WMO: 151180

Lat: 46.6899N

Lon: 22.6233E

Elev: 1109

StdP: 88.69

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -18.7	(c) -16.1	(d) -20.1	(e) 0.7	(f) -17.4	(g) -17.5	(h) 0.9	(i) -15.8	(j) 5.9	(k) -1.5	(l) 3.8	(m) -2.2	(n) 0.4	(o) 140	(p) N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 12.8	(c) 25.9	(d) 17.1	(e) 24.0	(f) 17.0	(g) 22.2	(h) 16.3	(i) 18.6	(j) 23.5	(k) 17.8	(l) 22.3	(m) 17.0	(n) 21.5	(o) 1.1	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.2	14.0	20.3	16.2	13.2	19.6	15.3	12.4	18.9	57.7	24.1	54.8	22.3	52.1	21.5	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 4.0	(b) 2.7	(c) 2.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
(a) 4.0	(b) 2.7	(c) 2.1	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-5.1	-2.4	-0.1	4.5	9.5	13.1	14.0	14.7	10.3	6.0	1.8	-2.8
	DBStd	7.88	4.93	4.53	3.98	4.64	3.32	3.27	2.87	2.97	4.09	4.20	4.69	4.55
	HDD10.0	2188	467	348	312	171	48	9	4	2	48	133	248	398
	HDD18.3	4753	725	582	571	416	273	159	135	116	241	382	497	656
	CDD10.0	487	0	0	0	5	33	101	129	149	58	9	1	0
	CDD18.3	9	0	0	0	0	0	2	2	4	1	0	0	0
	CDH23.3	238	0	0	0	0	4	26	77	119	12	0	0	0
	CDH26.7	31	0	0	0	0	1	0	7	23	1	0	0	0
Wind	WSAvg	0.6	0.5	0.6	0.6	0.7	0.6	0.6	0.6	0.5	0.5	0.5	0.6	0.5
Precipitation	PrecAvg	1058	66	58	69	81	101	126	125	84	98	85	79	86
	PrecMax	1437	140	157	200	129	175	211	233	227	271	224	177	194
	PrecMin	777	29	13	21	19	38	32	51	5	15	6	1	5
	PrecStd	173	29	34	47	31	37	44	55	48	56	51	36	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	9.9	14.1	20.5	23.2	25.6	27.6	29.0	25.0	20.3	15.1	7.8
		MCWB	3.8	5.0	7.0	11.8	14.6	17.5	17.2	18.7	16.4	12.9	9.1	5.2
	2%	DB	4.0	7.5	11.1	18.0	20.6	24.0	25.6	26.4	22.1	18.2	13.0	5.6
		MCWB	2.9	4.2	5.6	10.4	13.9	17.6	17.0	17.3	15.2	12.2	8.7	3.9
	5%	DB	2.6	5.6	8.5	15.5	18.6	22.3	23.2	23.8	20.0	15.7	10.8	4.0
		MCWB	1.8	3.2	4.5	9.3	12.8	16.8	16.7	17.0	14.4	11.3	7.8	2.7
	10%	DB	1.2	3.9	6.4	12.7	16.7	20.5	21.2	21.7	17.8	13.1	8.5	2.4
		MCWB	0.6	2.4	3.3	7.6	12.1	15.8	15.8	16.4	13.8	10.1	6.7	1.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.6	6.6	8.3	12.6	15.6	19.9	19.1	19.9	17.1	13.7	10.2	5.9
		MCDB	5.3	8.6	13.2	19.2	20.5	23.7	23.7	26.1	22.5	17.4	13.4	7.3
	2%	WB	3.1	4.6	6.3	11.1	14.5	18.4	18.2	18.7	16.0	12.9	9.1	4.5
		MCDB	3.7	6.7	10.3	17.4	19.1	22.5	22.8	24.3	20.8	16.9	12.3	5.6
	5%	WB	1.9	3.6	4.9	9.7	13.6	17.3	17.4	17.8	15.0	11.8	8.1	2.9
		MCDB	2.5	5.2	8.1	14.9	17.9	21.6	22.0	22.5	19.2	15.8	10.6	3.9
	10%	WB	0.7	2.4	3.6	8.1	12.5	16.1	16.5	17.0	14.0	10.4	7.1	1.8
		MCDB	1.1	3.7	6.0	11.8	16.2	20.0	20.8	21.4	17.5	12.9	8.8	2.5
Mean Daily Temperature Range	5% DB	MDBR	9.3	9.6	10.3	11.7	11.2	11.8	12.2	12.8	10.8	11.0	8.3	7.6
		MCDBR	9.5	11.4	13.9	16.5	14.9	14.5	16.2	16.5	14.7	15.6	11.5	8.1
		MCWBR	7.9	8.0	9.0	9.2	9.0	8.4	8.9	8.5	8.3	9.8	7.3	6.4
	5% WB	MCDBR	8.0	9.6	12.9	14.9	13.8	13.5	14.2	14.6	14.0	15.1	10.4	7.5
		MCWBR	6.7	7.0	8.8	8.7	8.7	8.5	8.4	8.3	8.3	9.8	7.0	6.1
Clear-Sky Solar Irradiance	taub	0.292	0.319	0.358	0.405	0.391	0.387	0.402	0.405	0.373	0.343	0.317	0.294	
	taud	2.362	2.313	2.260	2.155	2.251	2.290	2.249	2.237	2.307	2.387	2.407	2.399	
	Ebn at Noon	799	846	859	847	874	878	857	838	834	808	760	754	
	Edh at Noon	72	94	115	140	132	128	132	128	109	86	70	63	
All-Sky Solar Radiation	RadAvg	1.18	1.98	3.15	4.36	5.28	5.82	5.70	5.25	3.72	2.52	1.44	0.92	
	RadStd	0.17	0.26	0.46	0.46	0.51	0.47	0.50	0.50	0.43	0.30	0.20	0.12	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (64 neighbors)		+0.71	N/A	N/A	+1.00	N/A	N/A	-130	-197	+115	+58	

Nomenclature: See separate page